The given equation represents the **vector triple product identity**:

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$$

Example Calculation

Let's take specific vectors:

$$\mathbf{u} = (1, 2, 3), \quad \mathbf{v} = (4, 5, 6), \quad \mathbf{w} = (7, 8, 9)$$

Step 1: Compute $\mathbf{v} \times \mathbf{w}$

The cross product of two vectors $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$ is given by:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

For $\mathbf{v} = (4, 5, 6)$ and $\mathbf{w} = (7, 8, 9)$:

For $\mathbf{v} = (4, 5, 6)$ and $\mathbf{w} = (7, 8, 9)$:

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

Expanding along the first row:

$$= \mathbf{i}(5 \cdot 9 - 6 \cdot 8) - \mathbf{j}(4 \cdot 9 - 6 \cdot 7) + \mathbf{k}(4 \cdot 8 - 5 \cdot 7)$$

$$= \mathbf{i}(45 - 48) - \mathbf{j}(36 - 42) + \mathbf{k}(32 - 35)$$

$$= \mathbf{i}(-3) - \mathbf{j}(-6) + \mathbf{k}(-3)$$

$$= (-3, 6, -3)$$

Step 2: Compute $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$

Now, we compute:

$$\mathbf{u} \times (-3, 6, -3)$$

Using the determinant formula again:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ -3 & 6 & -3 \end{vmatrix}$$

Expanding along the first row:

$$= \mathbf{i}(2 \cdot (-3) - 3 \cdot 6) - \mathbf{j}(1 \cdot (-3) - 3 \cdot (-3)) + \mathbf{k}(1 \cdot 6 - 2 \cdot (-3))$$

$$= \mathbf{i}(-6 - 18) - \mathbf{j}(-3 + 9) + \mathbf{k}(6 + 6)$$

$$= \mathbf{i}(-24) - \mathbf{j}(6) + \mathbf{k}(12)$$

$$= (-24, -6, 12)$$

Step 3: Compute Right-Hand Side of Identity

We now compute:

$$(\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$$

1. Compute $\mathbf{u} \cdot \mathbf{w}$:

$$(1,2,3) \cdot (7,8,9) = 1(7) + 2(8) + 3(9) = 7 + 16 + 27 = 50$$

$$(\mathbf{u} \cdot \mathbf{w})\mathbf{v} = 50(4,5,6) = (200,250,300)$$

2. Compute $\mathbf{u} \cdot \mathbf{v}$:

$$(1,2,3) \cdot (4,5,6) = 1(4) + 2(5) + 3(6) = 4 + 10 + 18 = 32$$

$$(\mathbf{u} \cdot \mathbf{v})\mathbf{w} = 32(7,8,9) = (224,256,288)$$

3. Compute the final result:

$$(200,250,300) - (224,256,288) = (200 - 224, 250 - 256, 300 - 288)$$

$$= (-24, -6, 12)$$

Step 4: Verify Identity

Since both sides of the equation are equal:

$$(200, 250, 300) - (224, 256, 288) = (200 - 224, 250 - 256, 300 - 288)$$

$$= (-24, -6, 12)$$

Step 4: Verify Identity

Since both sides of the equation are equal:

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$$